

DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I

WEEK 5: LECTURE NOTE

QUADRATIC FUNCTIONS

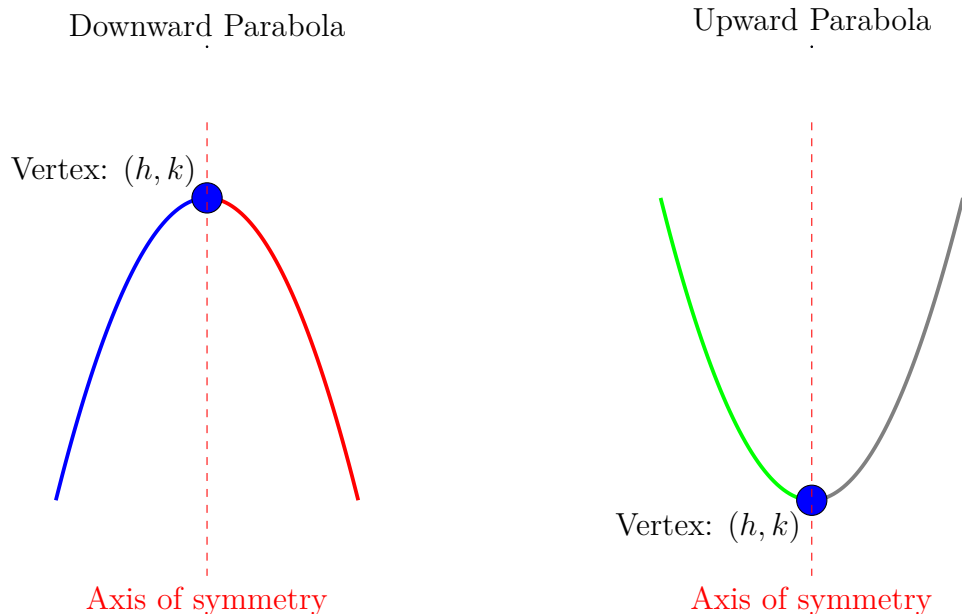
Definition 1 (Quadratic Function). Let a , b and c be real numbers with $a \neq 0$. The function given by

$$f(x) = ax^2 + bx + c$$

is called a **quadratic function**.

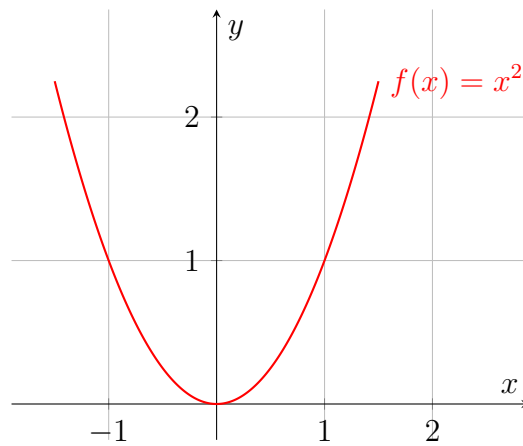
Properties of Quadratic Functions

1. The graph of a quadratic function is a special type of U-shaped curve called a **parabola** that opens upward or downward. Parabolas occur in many real-life applications, especially those involving reflective properties, such as satellite dishes or flashlight reflectors.
2. All parabolas are symmetric with respect to a line called **the axis of symmetry**, or simply **the axis of the parabola**.
3. The point where the axis intersects the parabola is called **the vertex of the parabola**.



- Domain of a Quadratic Function is $\mathbb{R} = (-\infty, \infty)$.
- Range of upward parabola is $[k, \infty)$.
- Range of downward parabola is $(-\infty, k]$.

Example 2 (THE PARENT QUADRATIC FUNCTION $f(x) = x^2$). The basic characteristics of **parent quadratic function** $f(x) = x^2$



Domain: $(-\infty, \infty)$

Range: $[0, \infty)$

Intercept: $(0, 0)$

Axis of symmetry: y -axis

Vertex: $(0, 0)$

Intercepts of the Graph of a Quadratic Function

Consider the general quadratic function

$$f(x) = ax^2 + bx + c, \quad a \neq 0.$$

Then obviously the y -intercept of the graph is the point $(0, c)$. To find x -intercepts (if any) we solve the quadratic equation

$$ax^2 + bx + c = 0.$$

We have the following methods to solve this equation:

1. Factorizing: Rewriting as $(x - p)(x - q) = 0$ if possible.
2. Completing the Square: Rewriting as $\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$
3. Quadratic Formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

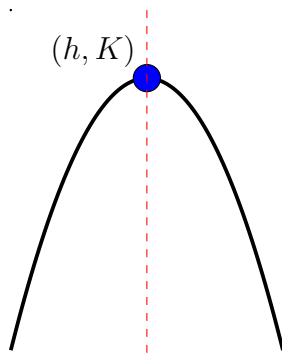
Note that it follows from Quadratic Formula that the graph has no x -intercepts in cases $b^2 - 4ac < 0$.

Definition 3 (Standard Form of a Quadratic Function). *The quadratic function given by*

$$f(x) = a(x - h)^2 + k$$

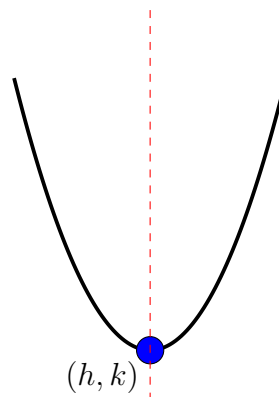
*is in standard form. The graph of f is a parabola whose axis is the vertical line $x = h$ and whose vertex is the point (h, k) . When $a > 0$ the parabola **opens upward**, and when $a < 0$, the parabola **opens downward**.*

$$f(x) = a(x - h)^2 + k, \quad a < 0$$



Axis of symmetry: $x = h$

$$f(x) = a(x - h)^2 + k, \quad a > 0$$



Axis of symmetry: $x = h$

Example 4 (Writing the Equation of a Parabola in Standard Form). Write the standard form of the equation of the parabola whose vertex is $(1, 2)$ and that passes through the point $(3, -6)$.

Solution. Because the vertex of the parabola is $(h, k) = (1, 2)$, substituting for h and k in standard form, the equation has the form $f(x) = a(x - 1)^2 + 2$. Because the parabola passes through the point $(3, -6)$, it follows that $f(3) = -6$. So, we obtain

$$f(x) = a(x - 1)^2 + 2$$

Write in standard form.

$$-6 = a(3 - 1)^2 + 2$$

Substitute -6 for $f(x)$ and 3 for x

$$-6 = 4a + 2$$

Simplify.

$$-8 = 4a$$

Subtract 2 from each side.

$$-2 = a$$

Divide each side by 4.

The equation in standard form is $\boxed{f(x) = -2(x - 1)^2 + 2}$.

By completing the square of the quadratic function $f(x) = ax^2 + bx + c$, we can rewrite the function in standard form.

$$f(x) = ax^2 + bx + c$$

Write Original Equation

$$= (ax^2 + bx) + c$$

Group x - terms

$$= a \left(x^2 + \frac{b}{a}x \right) + c$$

Factor a out of x - terms

$$= a \left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} \right) + c$$

Completing the Square

$$= a \left(x + \frac{b}{2a} \right)^2 + \left(c - \frac{b^2}{4a} \right)$$

Simplify

Therefore, the standard form is

$$f(x) = a \left(x + \frac{b}{2a} \right)^2 + \left(c - \frac{b^2}{4a} \right)$$

with vertex

$$\left(-\frac{b}{2a}, c - \frac{b^2}{4a} \right).$$

PROPERTIES OF THE GENERAL QUADRATIC FUNCTION

$$f(x) = ax^2 + bx + c$$

- The vertex of the graph of f is

$$\left(-\frac{b}{2a}, c - \frac{b^2}{4a}\right).$$

- If $a > 0$, then

(a) the parabola opens upward.

(b) the range of f is $\left[c - \frac{b^2}{4a}, \infty\right)$.

(c) f has the minimum value at $x = -\frac{b}{2a}$.

(d) The minimum value is

$$f\left(-\frac{b}{2a}\right) = c - \frac{b^2}{4a}.$$

- If $a < 0$, then

(a) the parabola opens downward.

(b) the range of f is

$$\left(-\infty, c - \frac{b^2}{4a}\right].$$

(c) f has the maximum value at $x = -\frac{b}{2a}$.

(d) The maximum value is

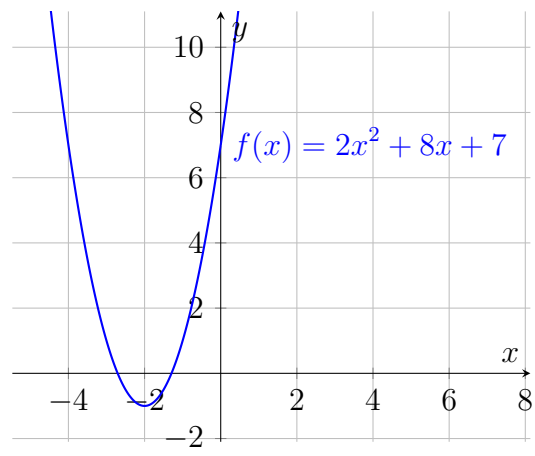
$$f\left(-\frac{b}{2a}\right) = c - \frac{b^2}{4a}.$$

Example 5 (Identifying the Vertex of a Quadratic Function). *Describe the graph of $f(x) = 2x^2 + 8x + 7$ and identify the vertex.*

Solution. *Write the quadratic function in standard form by completing the square. Recall that the first step is to factor out any coefficient of x^2 that is not 1.*

$f(x) = 2x^2 + 8x + 7$	Write original function.
$= (2x^2 + 8x) + 7$	Group x-terms.
$= 2(x^2 + 4x) + 7$	Factor 2 out of x-terms.
$= 2(x^2 + 4x + 2^2 - 2^2) + 7$	Complete the Square.
$= 2(x^2 + 4x + 4 - 4) + 7$	Simplify.
$= 2(x^2 + 4x + 4) - 8 + 7$	Regroup terms.
$= 2(x + 2)^2 - 1$	Write in Standard Form.

From the standard form, we can see that the graph of f is a parabola that opens upward with vertex $(-2, -1)$ as shown in the graph below.



Minimum value: $f(-2) = -1$
Axis of symmetry: $x = -2$